Table 8. Full sample comparison for $\delta = 0.05$, all small circuits test traces.

Trace	Size	Oracle	Inc-TVD	Inc-Hell	Inc-JS	Weissman	Hoeffding
dj_4_FEZ	4	100	**550**	2000	2000	2818	$8.27e+04$
dj_4_FEZ	4	150	**550**	2000	2000	2818	$8.27e+04$
dj_4_TOR	4	50	**550**	2300	3000	2818	$8.27e+04$
dj_4_TOR	4	300	**550**	2300	3000	2818	$8.27e+04$
dj_4_TOR	4	550	**550**	2300	3000	2818	$8.27e+04$
qaoa_4_SHE	4	300	**1150**	3700	3800	2818	$8.27e+04$
qaoa_4_SHE	4	1150	**1150**	3700	3800	2818	$8.27e+04$
qaoa_4_SHE	4	1600	**1150**	3700	3800	2818	$8.27e+04$
qaoa_6_MAR	6	1800	**3850**	6200	7300	9472	$1.61e+06$
qaoa_6_MAR	6	3000	**3850**	6200	7300	9472	$1.61e+06$
qaoa_6_MAR	6	3350	**3850**	6200	7300	9472	$1.61e+06$
qft_6_FEZ	6	3450	**4850**	6400	7300	9472	$1.61e+06$
qft_6_FEZ	6	3000	**4850**	6400	7300	9472	$1.61e+06$
qft_6_FEZ	6	3600	**4850**	6400	7300	9472	$1.61e+06$
qft_6_TOR	6	4000	**5350**	6600	7900	9472	$1.61e+06$
qft_6_TOR	6	3500	**5350**	6600	7900	9472	$1.61e+06$
qft_6_TOR	6	4250	**5350**	6600	7900	9472	$1.61e+06$
random_6_FEZ	6	2250	**4350**	6100	7400	9472	$1.61e+06$
random_6_FEZ	6	2700	**4350**	6100	7400	9472	$1.61e+06$
random_6_FEZ	6	3600	**4350**	6100	7400	9472	$1.61e+06$
vqe_6_KYI	6	150	**1350**	4800	5300	9472	$1.61e+06$
vqe_6_KYI	6	1500	**1350**	4800	5300	9472	$1.61e+06$
vqe_6_KYI	6	1550	**1350**	4800	5300	9472	$1.61e+06$
dj_8_FEZ	8	250	**1550**	5100	6200	$3.61e+04$	$3.03e+07$
dj_8_FEZ	8	3200	**1550**	5100	6200	$3.61e+04$	$3.03e+07$
dj_8_FEZ	8	3850	**1550**	5100	6200	$3.61e+04$	$3.03e+07$
dj_8_MAR	8	350	**1250**	6500	6500	$3.61e+04$	$3.03e+07$
dj_8_MAR	8	3050	**1250**	6500	6500	$3.61e+04$	$3.03e+07$
dj_8_MAR	8	3600	**1250**	6500	6500	$3.61e+04$	$3.03e+07$
qaoa_8_KYI	8	5850	**6550**	11200	14700	$3.61e+04$	$3.03e+07$
qaoa_8_KYI	8	8100	**6550**	11200	14700	$3.61e+04$	$3.03e+07$
qaoa_8_KYI	8	9350	**6550**	11200	14700	$3.61e+04$	$3.03e+07$
qaoa_8_SHE	8	5350	**6350**	12000	13900	$3.61e+04$	$3.03e+07$
qaoa_8_SHE	8	7850	**6350**	12000	13900	$3.61e+04$	$3.03e+07$
qaoa_8_SHE	8	9400	**6350**	12000	13900	$3.61e+04$	$3.03e+07$
qft_8_FEZ	8	8700	**9650**	12500	13700	$3.61e+04$	$3.03e+07$
qft_8_FEZ	8	7800	**9650**	12500	13700	$3.61e+04$	$3.03e+07$
qft_8_FEZ	8	9450	**9650**	12500	13700	$3.61e+04$	$3.03e+07$
qft_8_TOR	8	8450	**9850**	11700	13900	$3.61e+04$	$3.03e+07$
qft_8_TOR	8	7850	**9850**	11700	13900	$3.61e+04$	$3.03e+07$
qft_8_TOR	8	9450	**9850**	11700	13900	$3.61e+04$	$3.03e+07$
qnn_8_FEZ	8	7250	**8150**	12100	13600	$3.61e+04$	$3.03e+07$
qnn_8_FEZ	8	7450	**8150**	12100	13600	$3.61e+04$	$3.03e+07$
qnn_8_FEZ	8	9250	**8150**	12100	13600	$3.61e+04$	$3.03e+07$
qnn_8_KYI	8	8200	**9550**	11500	14100	$3.61e+04$	$3.03e+07$
qnn_8_KYI	8	7550	**9550**	11500	14100	$3.61e+04$	$3.03e+07$
qnn_8_KYI	8	9200	**9550**	11500	14100	$3.61e+04$	$3.03e+07$
random_8_FEZ	8	7600	**8250**	11500	14700	$3.61e+04$	$3.03e+07$
random_8_FEZ	8	8100	**8250**	11500	14700	$3.61e+04$	$3.03e+07$
random_8_FEZ	8	9500	**8250**	11500	14700	$3.61e+04$	$3.03e+07$
vqe_8_KYI	8	1000	**2450**	8400	8400	$3.61e+04$	$3.03e+07$
vqe_8_KYI	8	5700	**2450**	8400	8400	$3.61e+04$	$3.03e+07$
vqe_8_KYI	8	6700	**2450**	8400	8400	$3.61e+04$	$3.03e+07$

Table 9. Full sample comparison for $\delta = 0.05$, all large circuits test traces.

Trace	Size	Oracle	Inc-TVD	Inc-Hell	Inc-JS	Weissman	Hoeffding
dj_10_FEZ	10	500	**2550**	5600	9350	$1.43e+05$	$5.57e+08$
dj_10_FEZ	10	6750	**2550**	5600	9350	$1.43e+05$	$5.57e+08$
dj_10_FEZ	10	7750	**2550**	5600	9350	$1.43e+05$	$5.57e+08$
dj_10_TOR	10	1550	**3950**	9000	15250	$1.43e+05$	$5.57e+08$
dj_10_TOR	10	10050	**3950**	9000	15250	$1.43e+05$	$5.57e+08$
dj_10_TOR	10	10650	**3950**	9000	15250	$1.43e+05$	$5.57e+08$
qaoa_10_SHE	10	13800	**16650**	20000	20000	$1.43e+05$	$5.57e+08$
qaoa_10_SHE	10	15400	**16650**	20000	20000	$1.43e+05$	$5.57e+08$
qaoa_10_SHE	10	16350	**16650**	20000	20000	$1.43e+05$	$5.57e+08$
qaoa_12_MAR	12	14200	**15650**	20000	20000	$5.68e+05$	$1.01e+10$
qaoa_12_MAR	12	17550	**15650**	20000	20000	$5.68e+05$	$1.01e+10$
qaoa_12_MAR	12	17850	**15650**	20000	20000	$5.68e+05$	$1.01e+10$
qft_12_FEZ	12	18650	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
qft_12_FEZ	12	18550	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
qft_12_FEZ	12	18900	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
qft_12_TOR	12	18600	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
qft_12_TOR	12	18550	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
qft_12_TOR	12	18900	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
random_12_FEZ	12	18550	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
random_12_FEZ	12	18650	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
random_12_FEZ	12	18950	**20000**	20000	20000	$5.68e+05$	$1.01e+10$
vqe_12_KYI	12	3150	**5350**	20000	20000	$5.68e+05$	$1.01e+10$
vqe_12_KYI	12	13400	**5350**	20000	20000	$5.68e+05$	$1.01e+10$
vqe_12_KYI	12	14300	**5350**	20000	20000	$5.68e+05$	$1.01e+10$
dj_14_FEZ	14	2200	**3950**	12400	19550	$2.27e+06$	$1.80e+11$
dj_14_FEZ	14	12450	**3950**	12400	19550	$2.27e+06$	$1.80e+11$
dj_14_FEZ	14	12850	**3950**	12400	19550	$2.27e+06$	$1.80e+11$
dj_14_MAR	14	2350	**3550**	13200	14350	$2.27e+06$	$1.80e+11$
dj_14_MAR	14	10050	**3550**	13200	14350	$2.27e+06$	$1.80e+11$
dj_14_MAR	14	10650	**3550**	13200	14350	$2.27e+06$	$1.80e+11$
qaoa_14_KYI	14	17550	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qaoa_14_KYI	14	19350	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qaoa_14_KYI	14	19400	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qaoa_14_SHE	14	17900	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qaoa_14_SHE	14	19500	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qft_14_FEZ	14	18950	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qft_14_FEZ	14	19750	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qft_14_FEZ	14	19800	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qft_14_TOR	14	18950	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qft_14_TOR	14	19800	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qnn_14_FEZ	14	18950	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qnn_14_FEZ	14	19750	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qnn_14_FEZ	14	19800	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qnn_14_KYI	14	18950	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qnn_14_KYI	14	19750	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
qnn_14_KYI	14	19800	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
random_14_FEZ	14	18950	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
random_14_FEZ	14	19750	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
random_14_FEZ	14	19800	**20000**	20000	20000	$2.27e+06$	$1.80e+11$
vqe_14_KYI	14	8500	**7750**	20000	20000	$2.27e+06$	$1.80e+11$
vqe_14_KYI	14	17050	**7750**	20000	20000	$2.27e+06$	$1.80e+11$
vqe_14_KYI	14	17300	**7750**	20000	20000	$2.27e+06$	$1.80e+11$

Table 10. Full sample comparison for $\delta = 0.10$, all small circuits test traces.

Trace	Size	Oracle	Inc-TVD	Inc-Hell	Inc-JS	Weissman	Hoeffding
dj_4_FEZ	4	50	**150**	700	700	705	$2.07e+04$
dj_4_FEZ	4	100	**150**	700	700	705	$2.07e+04$
dj_4_TOR	4	50	**150**	550	500	705	$2.07e+04$
qaoa_4_SHE	4	150	**150**	1300	900	705	$2.07e+04$
qaoa_4_SHE	4	350	**150**	1300	900	705	$2.07e+04$
qaoa_4_SHE	4	400	**150**	1300	900	705	$2.07e+04$
qaoa_6_MAR	6	650	**1450**	1950	2400	2368	$4.02e+05$
qaoa_6_MAR	6	1350	**1450**	1950	2400	2368	$4.02e+05$
qaoa_6_MAR	6	1500	**1450**	1950	2400	2368	$4.02e+05$
qft_6_FEZ	6	1000	**1750**	1800	2500	2368	$4.02e+05$
qft_6_FEZ	6	850	**1750**	1800	2500	2368	$4.02e+05$
qft_6_FEZ	6	1050	**1750**	1800	2500	2368	$4.02e+05$
qft_6_TOR	6	850	2050	**1950**	2600	2368	$4.02e+05$
qft_6_TOR	6	800	2050	**1950**	2600	2368	$4.02e+05$
qft_6_TOR	6	1100	2050	**1950**	2600	2368	$4.02e+05$
random_6_FEZ	6	600	**1700**	1800	2600	2368	$4.02e+05$
random_6_FEZ	6	700	**1700**	1800	2600	2368	$4.02e+05$
random_6_FEZ	6	850	**1700**	1800	2600	2368	$4.02e+05$
vqe_6_KYI	6	50	**150**	1450	1500	2368	$4.02e+05$
vqe_6_KYI	6	300	**150**	1450	1500	2368	$4.02e+05$
dj_8_FEZ	8	100	**550**	2300	1900	9023	$7.56e+06$
dj_8_FEZ	8	600	**550**	2300	1900	9023	$7.56e+06$
dj_8_FEZ	8	950	**550**	2300	1900	9023	$7.56e+06$
dj_8_MAR	8	50	**150**	1400	2000	9023	$7.56e+06$
dj_8_MAR	8	700	**150**	1400	2000	9023	$7.56e+06$
qaoa_8_KYI	8	2000	**2350**	4150	5400	9023	$7.56e+06$
qaoa_8_KYI	8	3350	**2350**	4150	5400	9023	$7.56e+06$
qaoa_8_KYI	8	3950	**2350**	4150	5400	9023	$7.56e+06$
qaoa_8_SHE	8	1700	**2300**	4150	4600	9023	$7.56e+06$
qaoa_8_SHE	8	3350	**2300**	4150	4600	9023	$7.56e+06$
qaoa_8_SHE	8	3950	**2300**	4150	4600	9023	$7.56e+06$
qft_8_FEZ	8	3300	3650	**3300**	4900	9023	$7.56e+06$
qft_8_FEZ	8	2800	3650	**3300**	4900	9023	$7.56e+06$
qft_8_FEZ	8	3850	3650	**3300**	4900	9023	$7.56e+06$
qft_8_TOR	8	3600	3500	**3350**	4800	9023	$7.56e+06$
qft_8_TOR	8	3000	3500	**3350**	4800	9023	$7.56e+06$
qft_8_TOR	8	3950	3500	**3350**	4800	9023	$7.56e+06$
qnn_8_FEZ	8	2550	**3100**	3700	5000	9023	$7.56e+06$
qnn_8_FEZ	8	2950	**3100**	3700	5000	9023	$7.56e+06$
qnn_8_FEZ	8	3750	**3100**	3700	5000	9023	$7.56e+06$
qnn_8_KYI	8	3150	3550	**3500**	4500	9023	$7.56e+06$
qnn_8_KYI	8	3100	3550	**3500**	4500	9023	$7.56e+06$
qnn_8_KYI	8	3900	3550	**3500**	4500	9023	$7.56e+06$
random_8_FEZ	8	2750	**3100**	3600	5300	9023	$7.56e+06$
random_8_FEZ	8	3350	**3100**	3600	5300	9023	$7.56e+06$
random_8_FEZ	8	4150	**3100**	3600	5300	9023	$7.56e+06$
vqe_8_KYI	8	200	**1100**	2700	2700	9023	$7.56e+06$
vqe_8_KYI	8	750	**1100**	2700	2700	9023	$7.56e+06$
vqe_8_KYI	8	950	**1100**	2700	2700	9023	$7.56e+06$

Table 11. Full sample comparison for $\delta = 0.10$, all large circuits test traces.

Trace	Size	Oracle	Inc-TVD	Inc-Hell	Inc-JS	Weissman	Hoeffding
dj_10_FEZ	10	100	**1600**	3450	4200	$3.56e+04$	$1.39e+08$
dj_10_FEZ	10	1600	**1600**	3450	4200	$3.56e+04$	$1.39e+08$
dj_10_FEZ	10	1750	**1600**	3450	4200	$3.56e+04$	$1.39e+08$
dj_10_TOR	10	500	**2200**	4450	4700	$3.56e+04$	$1.39e+08$
dj_10_TOR	10	2400	**2200**	4450	4700	$3.56e+04$	$1.39e+08$
dj_10_TOR	10	2800	**2200**	4450	4700	$3.56e+04$	$1.39e+08$
qaoa_10_SHE	10	6550	9950	**9750**	11000	$3.56e+04$	$1.39e+08$
qaoa_10_SHE	10	9150	9950	**9750**	11000	$3.56e+04$	$1.39e+08$
qaoa_10_SHE	10	10300	9950	**9750**	11000	$3.56e+04$	$1.39e+08$
qaoa_12_MAR	12	7200	**9000**	14950	15900	$1.42e+05$	$2.52e+09$
qaoa_12_MAR	12	13250	**9000**	14950	15900	$1.42e+05$	$2.52e+09$
qaoa_12_MAR	12	13750	**9000**	14950	15900	$1.42e+05$	$2.52e+09$
qft_12_FEZ	12	15400	**17150**	18150	20000	$1.42e+05$	$2.52e+09$
qft_12_FEZ	12	15600	**17150**	18150	20000	$1.42e+05$	$2.52e+09$
qft_12_FEZ	12	16450	**17150**	18150	20000	$1.42e+05$	$2.52e+09$
qft_12_TOR	12	15200	**17200**	19350	20000	$1.42e+05$	$2.52e+09$
qft_12_TOR	12	15550	**17200**	19350	20000	$1.42e+05$	$2.52e+09$
qft_12_TOR	12	16550	**17200**	19350	20000	$1.42e+05$	$2.52e+09$
random_12_FEZ	12	15200	**17100**	19250	20000	$1.42e+05$	$2.52e+09$
random_12_FEZ	12	15700	**17100**	19250	20000	$1.42e+05$	$2.52e+09$
random_12_FEZ	12	16600	**17100**	19250	20000	$1.42e+05$	$2.52e+09$
vqe_12_KYI	12	550	**3400**	6150	7500	$1.42e+05$	$2.52e+09$
vqe_12_KYI	12	5250	**3400**	6150	7500	$1.42e+05$	$2.52e+09$
vqe_12_KYI	12	6200	**3400**	6150	7500	$1.42e+05$	$2.52e+09$
dj_14_FEZ	14	500	**2450**	4850	5200	$5.68e+05$	$4.49e+10$
dj_14_FEZ	14	4150	**2450**	4850	5200	$5.68e+05$	$4.49e+10$
dj_14_FEZ	14	4700	**2450**	4850	5200	$5.68e+05$	$4.49e+10$
dj_14_MAR	14	600	**2300**	5150	6200	$5.68e+05$	$4.49e+10$
dj_14_MAR	14	3100	**2300**	5150	6200	$5.68e+05$	$4.49e+10$
dj_14_MAR	14	3750	**2300**	5150	6200	$5.68e+05$	$4.49e+10$
qaoa_14_KYI	14	13350	**13400**	20000	20000	$5.68e+05$	$4.49e+10$
qaoa_14_KYI	14	17400	**13400**	20000	20000	$5.68e+05$	$4.49e+10$
qaoa_14_KYI	14	17650	**13400**	20000	20000	$5.68e+05$	$4.49e+10$
qaoa_14_SHE	14	14350	**14600**	20000	20000	$5.68e+05$	$4.49e+10$
qaoa_14_SHE	14	17750	**14600**	20000	20000	$5.68e+05$	$4.49e+10$
qaoa_14_SHE	14	17950	**14600**	20000	20000	$5.68e+05$	$4.49e+10$
qft_14_FEZ	14	17750	**17800**	20000	20000	$5.68e+05$	$4.49e+10$
qft_14_FEZ	14	19050	**17800**	20000	20000	$5.68e+05$	$4.49e+10$
qft_14_FEZ	14	19150	**17800**	20000	20000	$5.68e+05$	$4.49e+10$
qft_14_TOR	14	17750	**17750**	20000	20000	$5.68e+05$	$4.49e+10$
qft_14_TOR	14	19000	**17750**	20000	20000	$5.68e+05$	$4.49e+10$
qft_14_TOR	14	19100	**17750**	20000	20000	$5.68e+05$	$4.49e+10$
qnn_14_FEZ	14	17600	**17700**	20000	20000	$5.68e+05$	$4.49e+10$
qnn_14_FEZ	14	19000	**17700**	20000	20000	$5.68e+05$	$4.49e+10$
qnn_14_FEZ	14	19050	**17700**	20000	20000	$5.68e+05$	$4.49e+10$
qnn_14_KYI	14	17700	**17750**	20000	20000	$5.68e+05$	$4.49e+10$
qnn_14_KYI	14	19000	**17750**	20000	20000	$5.68e+05$	$4.49e+10$
qnn_14_KYI	14	19100	**17750**	20000	20000	$5.68e+05$	$4.49e+10$
random_14_FEZ	14	17700	**17800**	20000	20000	$5.68e+05$	$4.49e+10$
random_14_FEZ	14	18950	**17800**	20000	20000	$5.68e+05$	$4.49e+10$
random_14_FEZ	14	19050	**17800**	20000	20000	$5.68e+05$	$4.49e+10$
vqe_14_KYI	14	2600	**4500**	11150	14000	$5.68e+05$	$4.49e+10$
vqe_14_KYI	14	11400	**4500**	11150	14000	$5.68e+05$	$4.49e+10$
vqe_14_KYI	14	12050	**4500**	11150	14000	$5.68e+05$	$4.49e+10$

Table 12. Full sample comparison for $\delta = 0.25$, all small circuits test traces.

Trace	Size	Oracle	Inc-TVD	Inc-Hell	Inc-JS	Weissman	Hoeffding
dj_4_FEZ	4	50	**100**	150	250	113	3309
dj_4_TOR	4	50	**100**	200	250	113	3309
qaoa_4_SHE	4	50	**250**	400	250	113	3309
qaoa_6_MAR	6	100	**300**	350	450	379	$6.43e+04$
qaoa_6_MAR	6	150	**300**	350	450	379	$6.43e+04$
qft_6_FEZ	6	250	500	**450**	750	379	$6.43e+04$
qft_6_FEZ	6	300	500	**450**	750	379	$6.43e+04$
qft_6_TOR	6	200	**500**	500	750	379	$6.43e+04$
qft_6_TOR	6	250	**500**	500	750	379	$6.43e+04$
random_6_FEZ	6	150	**400**	400	650	379	$6.43e+04$
random_6_FEZ	6	250	**400**	400	650	379	$6.43e+04$
vqe_6_KYI	6	50	**150**	250	250	379	$6.43e+04$
dj_8_FEZ	8	50	**150**	250	250	1444	$1.21e+06$
dj_8_MAR	8	50	**100**	300	250	1444	$1.21e+06$
qaoa_8_KYI	8	300	**550**	750	850	1444	$1.21e+06$
qaoa_8_KYI	8	700	**550**	750	850	1444	$1.21e+06$
qaoa_8_KYI	8	850	**550**	750	850	1444	$1.21e+06$
qaoa_8_SHE	8	350	**500**	850	850	1444	$1.21e+06$
qaoa_8_SHE	8	600	**500**	850	850	1444	$1.21e+06$
qaoa_8_SHE	8	750	**500**	850	850	1444	$1.21e+06$
qft_8_FEZ	8	650	**850**	850	1050	1444	$1.21e+06$
qft_8_FEZ	8	800	**850**	850	1050	1444	$1.21e+06$
qft_8_FEZ	8	900	**850**	850	1050	1444	$1.21e+06$
qft_8_TOR	8	650	**850**	1100	1050	1444	$1.21e+06$
qft_8_TOR	8	750	**850**	1100	1050	1444	$1.21e+06$
qft_8_TOR	8	900	**850**	1100	1050	1444	$1.21e+06$
qnn_8_FEZ	8	500	**700**	1050	950	1444	$1.21e+06$
qnn_8_FEZ	8	750	**700**	1050	950	1444	$1.21e+06$
qnn_8_FEZ	8	900	**700**	1050	950	1444	$1.21e+06$
qnn_8_KYI	8	650	**800**	1050	1050	1444	$1.21e+06$
qnn_8_KYI	8	750	**800**	1050	1050	1444	$1.21e+06$
qnn_8_KYI	8	900	**800**	1050	1050	1444	$1.21e+06$
random_8_FEZ	8	550	**700**	1000	950	1444	$1.21e+06$
random_8_FEZ	8	800	**700**	1000	950	1444	$1.21e+06$
random_8_FEZ	8	950	**700**	1000	950	1444	$1.21e+06$
vqe_8_KYI	8	50	**200**	500	450	1444	$1.21e+06$
vqe_8_KYI	8	150	**200**	500	450	1444	$1.21e+06$

Table 13. Full sample comparison for $\delta = 0.25$, all large circuits test traces.

Trace	Size	Oracle	Inc-TVD	Inc-Hell	Inc-JS	Weissman	Hoeffding
dj_10_FEZ	10	50	**300**	1200	1550	5703	$2.23e+07$
dj_10_FEZ	10	100	**300**	1200	1550	5703	$2.23e+07$
dj_10_TOR	10	50	1800	1700	**1650**	5703	$2.23e+07$
dj_10_TOR	10	250	1800	1700	**1650**	5703	$2.23e+07$
dj_10_TOR	10	300	1800	1700	**1650**	5703	$2.23e+07$
qaoa_10_SHE	10	1300	6900	**3400**	4100	5703	$2.23e+07$
qaoa_10_SHE	10	2400	6900	**3400**	4100	5703	$2.23e+07$
qaoa_10_SHE	10	2650	6900	**3400**	4100	5703	$2.23e+07$
qaoa_12_MAR	12	1500	6300	**5000**	5900	$2.27e+04$	$4.03e+08$
qaoa_12_MAR	12	3800	6300	**5000**	5900	$2.27e+04$	$4.03e+08$
qaoa_12_MAR	12	4250	6300	**5000**	5900	$2.27e+04$	$4.03e+08$
qft_12_FEZ	12	6800	13300	**9200**	10100	$2.27e+04$	$4.03e+08$
qft_12_FEZ	12	8400	13300	**9200**	10100	$2.27e+04$	$4.03e+08$
qft_12_FEZ	12	9050	13300	**9200**	10100	$2.27e+04$	$4.03e+08$
qft_12_TOR	12	6650	13300	9500	**9450**	$2.27e+04$	$4.03e+08$
qft_12_TOR	12	8250	13300	9500	**9450**	$2.27e+04$	$4.03e+08$
qft_12_TOR	12	8950	13300	9500	**9450**	$2.27e+04$	$4.03e+08$
random_12_FEZ	12	6650	13100	**9200**	9900	$2.27e+04$	$4.03e+08$
random_12_FEZ	12	8450	13100	**9200**	9900	$2.27e+04$	$4.03e+08$
random_12_FEZ	12	9100	13100	**9200**	9900	$2.27e+04$	$4.03e+08$
vqe_12_KYI	12	50	2600	**2400**	3100	$2.27e+04$	$4.03e+08$
vqe_12_KYI	12	400	2600	**2400**	3100	$2.27e+04$	$4.03e+08$
vqe_12_KYI	12	500	2600	**2400**	3100	$2.27e+04$	$4.03e+08$
dj_14_FEZ	14	50	1900	**1800**	2250	$9.09e+04$	$7.19e+09$
dj_14_FEZ	14	200	1900	**1800**	2250	$9.09e+04$	$7.19e+09$
dj_14_FEZ	14	250	1900	**1800**	2250	$9.09e+04$	$7.19e+09$
dj_14_MAR	14	100	1900	**1800**	2450	$9.09e+04$	$7.19e+09$
dj_14_MAR	14	200	1900	**1800**	2450	$9.09e+04$	$7.19e+09$
dj_14_MAR	14	300	1900	**1800**	2450	$9.09e+04$	$7.19e+09$
qaoa_14_KYI	14	4700	9100	**8800**	9200	$9.09e+04$	$7.19e+09$
qaoa_14_KYI	14	8700	9100	**8800**	9200	$9.09e+04$	$7.19e+09$
qaoa_14_KYI	14	9350	9100	**8800**	9200	$9.09e+04$	$7.19e+09$
qaoa_14_SHE	14	5950	10100	**8200**	10450	$9.09e+04$	$7.19e+09$
qaoa_14_SHE	14	9750	10100	**8200**	10450	$9.09e+04$	$7.19e+09$
qaoa_14_SHE	14	10250	10100	**8200**	10450	$9.09e+04$	$7.19e+09$
qft_14_FEZ	14	12700	**14200**	17900	19350	$9.09e+04$	$7.19e+09$
qft_14_FEZ	14	15000	**14200**	17900	19350	$9.09e+04$	$7.19e+09$
qft_14_FEZ	14	15300	**14200**	17900	19350	$9.09e+04$	$7.19e+09$
qft_14_TOR	14	12650	**14200**	17800	18850	$9.09e+04$	$7.19e+09$
qft_14_TOR	14	14950	**14200**	17800	18850	$9.09e+04$	$7.19e+09$
qft_14_TOR	14	15250	**14200**	17800	18850	$9.09e+04$	$7.19e+09$
qnn_14_FEZ	14	12100	**14100**	17000	18400	$9.09e+04$	$7.19e+09$
qnn_14_FEZ	14	14600	**14100**	17000	18400	$9.09e+04$	$7.19e+09$
qnn_14_FEZ	14	14950	**14100**	17000	18400	$9.09e+04$	$7.19e+09$
qnn_14_KYI	14	12550	**14200**	17700	19350	$9.09e+04$	$7.19e+09$
qnn_14_KYI	14	14900	**14200**	17700	19350	$9.09e+04$	$7.19e+09$
qnn_14_KYI	14	15200	**14200**	17700	19350	$9.09e+04$	$7.19e+09$
random_14_FEZ	14	12500	**14300**	17700	19100	$9.09e+04$	$7.19e+09$
random_14_FEZ	14	14850	**14300**	17700	19100	$9.09e+04$	$7.19e+09$
random_14_FEZ	14	15150	**14300**	17700	19100	$9.09e+04$	$7.19e+09$
vqe_14_KYI	14	450	3500	**3400**	3550	$9.09e+04$	$7.19e+09$
vqe_14_KYI	14	1800	3500	**3400**	3550	$9.09e+04$	$7.19e+09$
vqe_14_KYI	14	2100	3500	**3400**	3550	$9.09e+04$	$7.19e+09$

Received 20 February 2007; revised 12 March 2009; accepted 5 June 2009